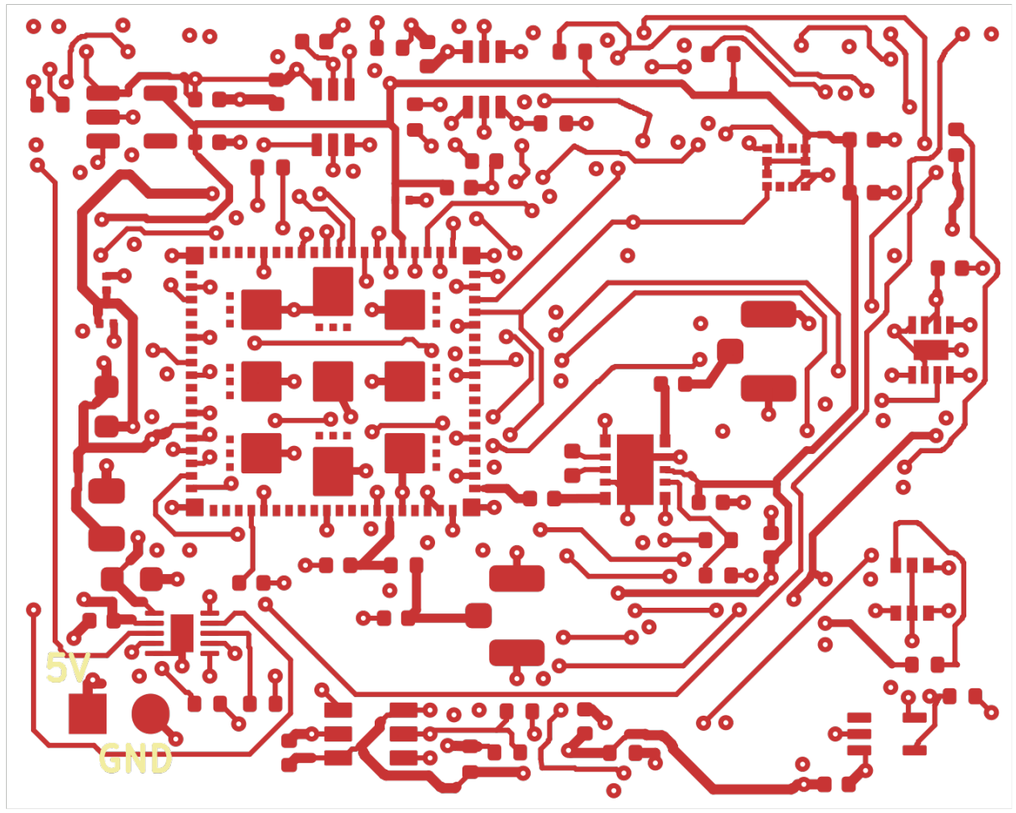
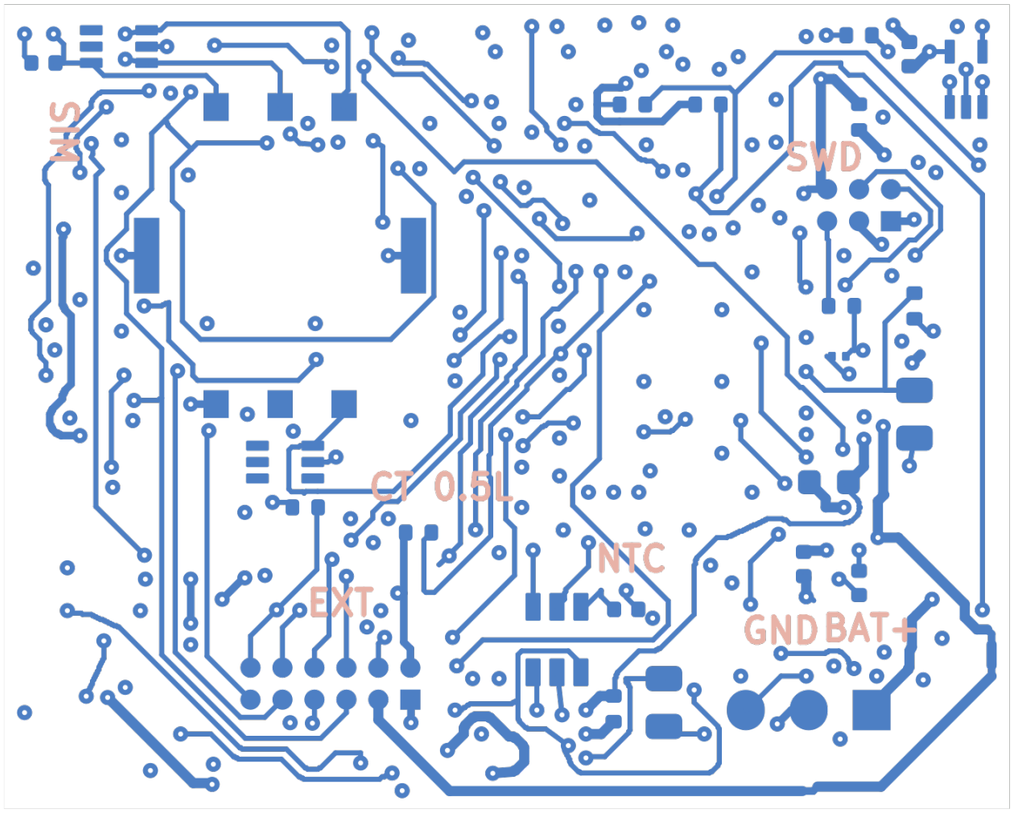
COLLAR TRACKER / VERIFIED PCB

r05l PCB DFM is complete. A fully assembled, working tracker is not released.

TOP / 40 x 32mm



BOTTOM / mirrored view



VERIFIED / UNCHANGED

Six copper layers; physical nano-SIM.

Native DRC / ERC / opens / parity: zero findings.

Both JLC PCB runs: 0 Danger / 0 Warning / 455 Good.

Same ZIP and rules; all 29 rows analyzed; SMT off.

81 fitted references / 44 candidate MPNs.

109 physical GND pads have local via paths.

Independent hardware hot/cold charge interlock.

REMAINING BEFORE A WORKING-DEVICE ORDER

Protected pack and lead-exit approval.

Dock holding and physical wire-clamp acceptance.

Internal J3 wire routes, strain relief and joining process.

Factory sourcing, CPL rotation and SMT review.

Firmware, programming and functional tests.

Powered / RF / physical water and dust tests.

No purchase, supplier message or physical test made.

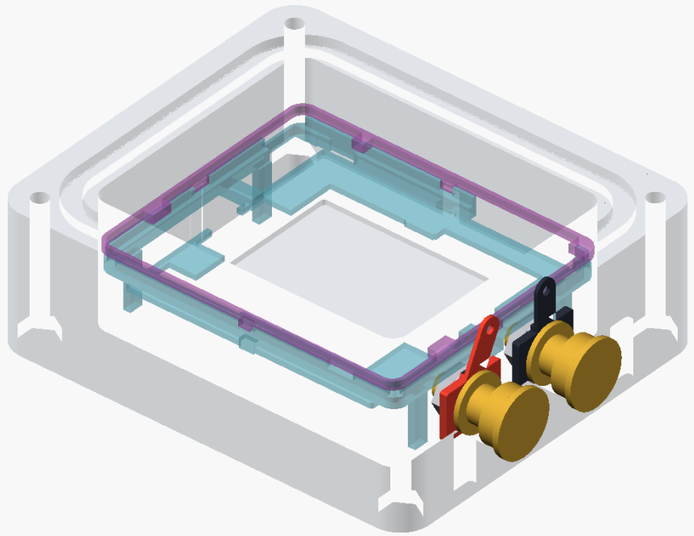
Gerber SHA256: c736c5d35ddfe4c4662e57606120f00f7358492e11d5fbb32eaf1d0ff6caf5d2

JLC records: 637366592129687553 / 637366801595977730

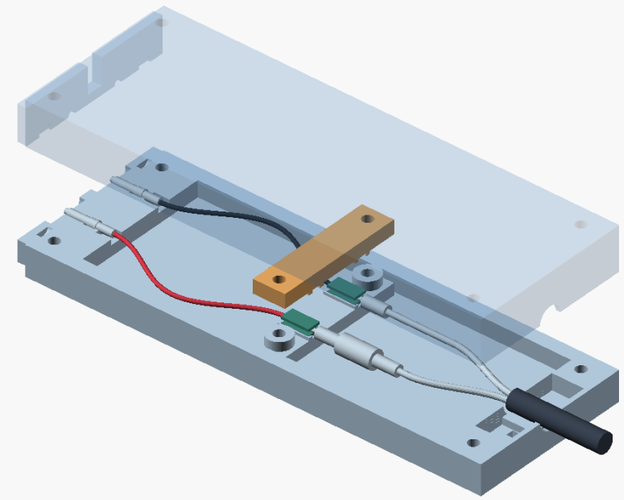
COLLAR TRACKER / CURRENT ASSEMBLY STUDY

Conditional geometry. Pack supplier response, factory assembly and physical validation remain open.

r16 / contact bearing plates and internal pockets



r15 dry dock with separate wire clamp



CURRENT MODEL

Rigid shell + lid: 57.4 x 49.4 x 21.0mm.

Body including external key: 61.4mm in X.

Boot: 60.4 x 52.4 x 25.8mm.

r12 carrier/cap; r11 lid, frame, boot and seals.

r16 shell/contacts; r15 dock: 101.5 x 36 x 16mm.

Constant-length pigtails; continuous radius >=8.696mm.

Separate padded clamp and factory forming fixture.

LIMITS / DO NOT ORDER AS A FINISHED DEVICE

Requested pack: 32 x 30.5 x 5.3mm maximum.

That pack envelope is not supplier approved.

Free-wire geometry does not constrain real buckling.

Contact holding, clamp pull and cycle tests remain open.

Charging is a dry indoor prototype only.

IP68 is a target, not a certified or measured result.

2m fresh water / 60min is an engineering assumption.

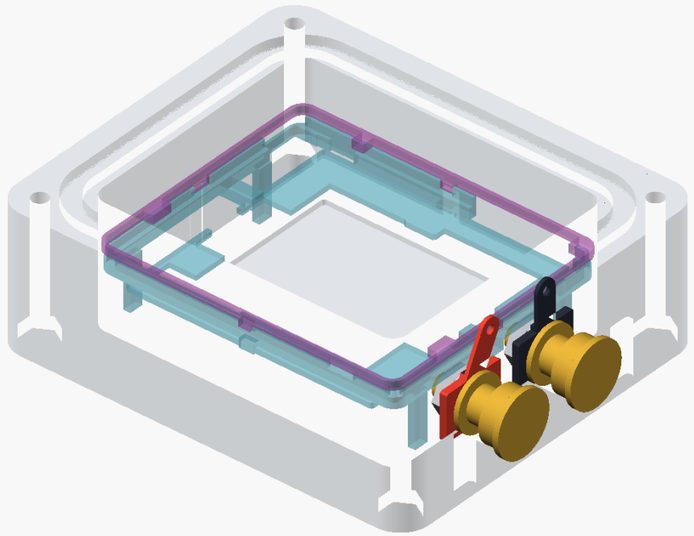
Parts in this review are explicitly revision-mapped; previous releases remain preserved.

Open the repository STATUS / RESUME for the exact source, commands and remaining work.

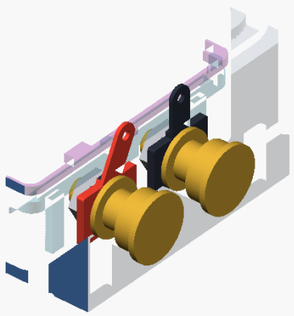
CT-CONTACT-RET-001-A / REAR CONTACT RETENTION

Geometric prototype. Internal wire routing, factory process and physical qualification remain open.

r16 / unchanged outside size; new dry-side bearing pockets



r16 / custom terminals, washer and M3 thin nut



CHANGED / DIMENSIONS IN mm

Bearing faces: case X=22.60 +/-0.05.

Contact stem: 5.90 +/-0.05; M3 x 0.5 thread: 4.80.

Terminal base: 9.0 wide; Z=2.35 to 8.65; 1.0 thick.

Tail: 0.50 thick above Z=9.25; 1.20 wire hole.

Positive tail: Y=-3.30, Z=13.30; red to J3.1.

Ground tail: Y=+6.00, Z=13.30; black to J3.2.

Washer HRDW-M3-A2 / nut HFN-M3-A2 candidates.

CHECKED / NOT A FINISHED CHARGING PATH

Old 6.4mm circular stack is smaller than the bore.

New plate bears on both sides of the 7.34mm bore.

Minimum calculated ear bearing: 4.413mm2 each.

All 85 swept populated envelopes clear new metal.

Minimum component gap: 2.146mm at C10.

Head, seal glands and dock contact locations retained.

PCB/Gerber unchanged: both actual JLC runs 0/0/455.

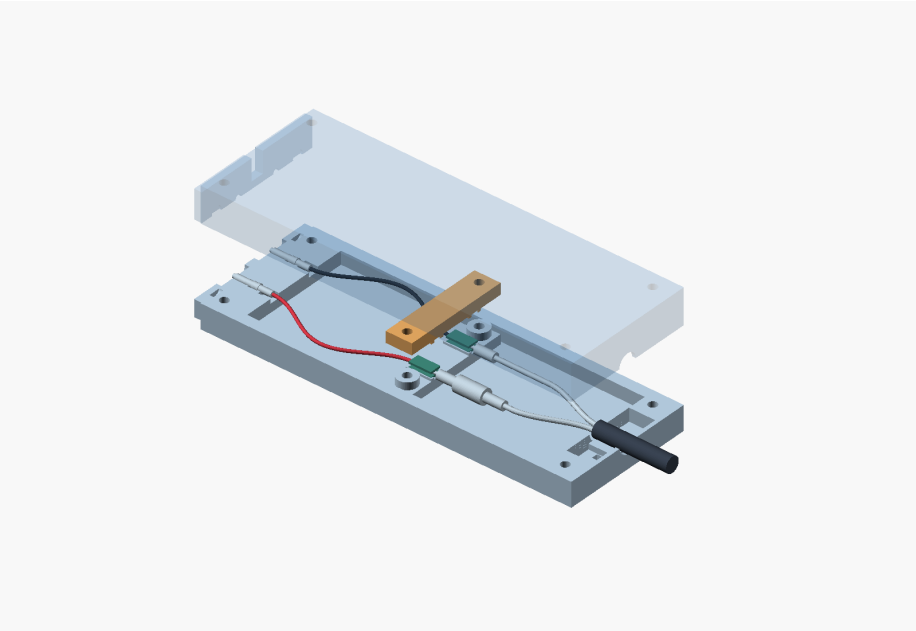
Custom machined terminals; use contact.json and contacts.scad. Not a purchased Keystone lug or a plastic printed part.

Torque, pull, resistance, plating, creep and water/dust tests remain open. No supplier enquiry or order was sent.

CT-DOCK-001-C | Wire clamp and assembly fixture / r15

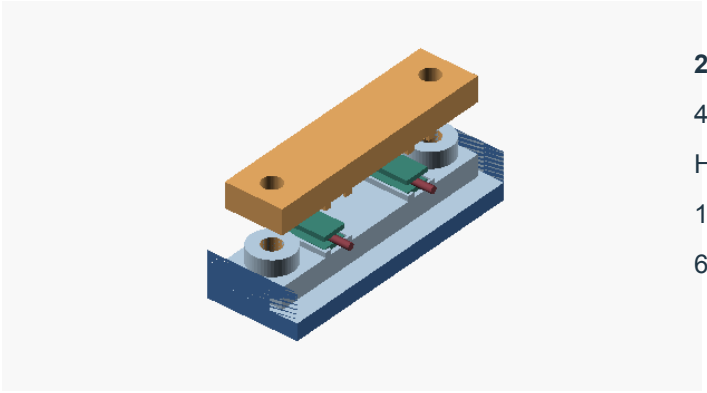
PROTOTYPE — geometry checked; clamp retention, dock holding and electrical release remain open

External dry dock: 101.5 x 36 x 16mm



Wearable, PCB, key, seals and spring-contact seats unchanged.

Clamp assembly (cap lifted for visibility)



2 x M2 x 10

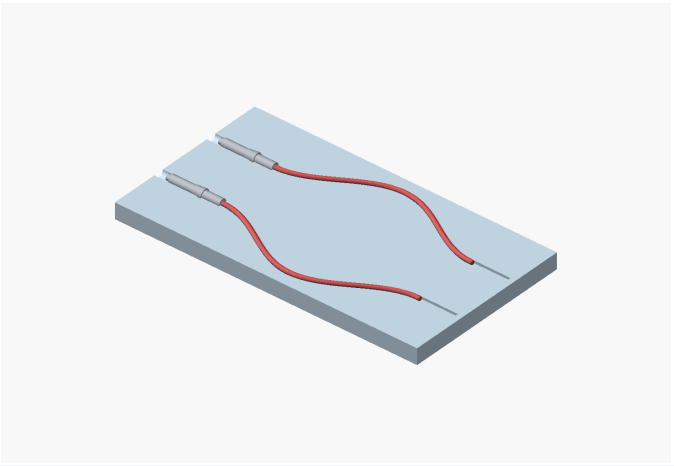
4 silicone pads

Hard-stop columns

1.93mm face gap

6 x 3mm pad outline

Separate factory forming fixture



Free span / wire	40.95 +/- 0.10mm at 23C
Fixed-end plane	X=49.00 +/- 0.10mm total
Sheet thickness	0.7874 +/- 0.1016mm
Local pad displacement	5.50..43.52% / assigned bounds
Housing / cap / fixture	Four closed CAD meshes
CAD collision checks	5 clear / 3 contacts / 5 controls
PCB DFM	Unchanged: two runs at 0/0/455

The fixture controls assembly shape only. Free wires still require cycle tests.

Squeeze is not a friction, insulation-strain or lifetime calculation.

Factory pull target: 5N / 60s per wire; no physical result exists.

Position error and later slip share the same +/- 0.10mm budget.

Open: latch, internal J3 route, pack approval, firmware and physical power/RF/water/dust testing. No order release.